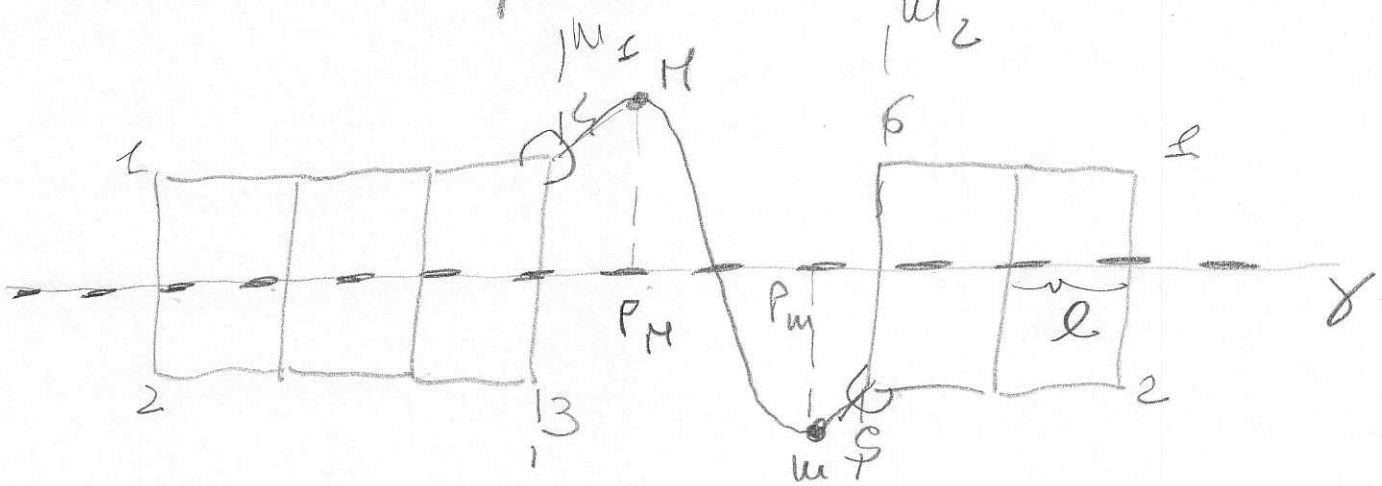
981 - 1519

Consider a net consisting of $3 \leq N \leq 5$ convex quadrilaterals with a diagram as in the picture



We draw the equator γ passing through the midpoints of the edges 12, 43 and 65 and the meridians M_1 and M_2 because the edge 45 joins an angle of $\frac{\pi}{6}$ in M_1 it goes up in latitude before reaching 5. Analogously from 5 it goes down.

If we denote by M and m

the points of maximal and minimal latitude in the edge 45, these must be antipodal. If we project them on the equator we also find two antipodal points. So if l is the length of the intersection of one quadrilateral with δ, we must have

$$Nl < \pi$$

but $l \geq \frac{2\pi}{3}$, so $N < 3$